



Technical Service Bulletin: TSB101088

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Crankshaft and Accessory Drive Seals

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This document introduces an updated list for the rotating shaft oil seals, dust seals (POSE), and installation/service tools that are available for the engines listed in the following tables.

Definition of Terms Used in this Topic

Production Seal

The production seal is the base seal used in new production engine build. In production, mounting parts are structured with the seal for installation on the assembly line. The mounting parts can include o-rings, gaskets, capscrews, and so forth. A production seal expander (a special tool, **not** a plastic or metal ring) is also used when the seal is installed over the shaft. This prevents damage to the sealing element during installation.

In cases where mounting parts or seal expanders are **not** required (pressed-in bore seals/shaft installed through the seal), the production seal is also released for service.

Service Seal with Installation Sleeve

The service seal includes a disposable sleeve for installations where the seal **must** be installed over the shaft. The service seal is shipped with the seal installed on the installation sleeve. The seal **must not** be removed from the installation sleeve prior to seal installation. During installation, the seal and sleeve are positioned on the end of the shaft as an assembly. The seal is then pushed off the installation sleeve and onto the shaft. The installation sleeve is then removed, and the appropriate seal installation tool is used to push the seal into the seal bore or to seat the seal against the housing, depending on the application.

Service Seal Assembly with Wear Sleeve

The service seal with wear sleeve was developed for use where the shaft sealing surface has been severely worn or damaged. The wear sleeve is 1.57-mm [0.062-in] thick to provide a robust wear surface for the seal. The seal used with the wear sleeve was designed specifically for use with the 1.57-mm [0.062-in] thick wear sleeve. The seal is **not** interchangeable with the standard

production or service seal since it has a larger inside diameter and fits **only** the thick-wall wear sleeve. If either the wear sleeve or seal is replaced, or the seal is removed, the complete seal and wear sleeve assembly **must** be replaced.

The press-fit wear sleeve and seal assembly **must** be installed in one operation. The installation requires the use of a special service tool that was designed specifically to install the seal assembly. Refer to the engine model and seal position in this topic to determine the service tool required to install the service seal with wear sleeve. The assembly **can not** be installed on the shaft without the special installation service tool.

The service seal with wear sleeve assembly is **not** available for all engine models or seal positions included in this topic. **Only** those listed in this topic under each specific engine model and seal position are available at this time.

Positive-on-Shaft Excluder (POSE)

The POSE seal was developed to improve primary seal durability. The POSE seal is installed in front of the primary seal and rotates with the shaft. The POSE seal was developed to improve primary seal durability. The POSE seal is installed in front of the primary seal and rotates with the shaft, which reduces the amount of dirt reaching the primary sealing element. This significantly increases seal life, especially in applications in which the seal is exposed to a highly abrasive environment. The POSE seal was designed to be used in conjunction with standard oil seals. No special service tools are required to install the POSE seal. The POSE seal can be installed onto the shaft using **only** hand-pressure.

The POSE seal is **not** available for all engine models or seal positions included in this topic. **Only** those listed in this topic under each specific engine model and seal position are available at this time.

Service Tool for Service Seal Installation

The service tools developed for installation of the standard service seal in some cases are multipurpose, and can also be used to remove the seal. The tools are designed to install the service seal properly to the correct depth in the seal bore without damaging the sealing element, shaft, or seal bore. It is recommended that the proper service tool be used anytime seal replacement is required. Attempting to install the service seal without the proper installation tool can result in seal, shaft, or bore damage that will progress into oil leaks.

Service Tool for Installation of Service Seal with Wear Sleeve

This service tool is designed specifically for installation of the service seal with wear sleeve. The service seal and 1.57-mm [0.060-in] thick wear sleeve are available **only** as an assembly and can be installed **only** by using the proper service tool. The sleeve and wear sleeve assembly **must not** be disassembled for any reason during the installation process. Removal of the seal from the wear sleeve will result in seal damage; the seal can **not** be reinstalled on the wear sleeve without damaging the sealing lip.

Do **not** attempt to install the seal/wear sleeve without the proper service tool.

Repair Sleeve 0.304-mm [0.012-in thick]

The 0.304-mm [0.012-in] thick repair sleeve, commonly referred to as the “speedi sleeve,” has been used with limited success for several years. The repair sleeve was designed for use on rotating shafts that had experienced minor wear in the seal contact area. A standard service seal is used in conjunction with the repair sleeve. The use of a standard service seal on the thin-walled repair sleeve increases seal-lip loading, which occasionally results in seal failure because of extreme temperatures generated by the friction between the seal lip and the repair sleeve.

Installation of the thin-wall wear sleeve is also extremely critical because of the wall's thickness. Any distortion of the repair sleeve during installation will result in oil leakage between the sleeve and shaft. The working environment of the engine **must** also be considered when using the repair sleeve. The service life of the repair sleeve is extremely limited in highly abrasive environments because of the wall thickness of the repair sleeve.

The 0.034-mm [0.012-in] thick repair sleeve is considered a temporary repair **only** and **must** be used with discretion. Cummins has obsoleted some of these repair sleeves because of poor performance in the field. These thin-walled repair sleeves are being replaced with the service seal wear sleeve.

Unitized Seals

Unitized (one-piece) accessory drive and crankshaft seals were introduced with the release of the QSK19 engine. The unitized seal kits released for service of the QSK19 contain a disposable plastic installation tool. The tool **must** be used to install the service seal. The seal is **not** available for service without the tool.

The unitized seals are also used on the QSK45/60 engines; however, the installation tool is **not** included with the service seal. The accessory drive seals do **not** require an installation tool. The seal is pressed into position during installation of the accessory drive pulley. The front and rear crankshaft seal do require an installation tool. Reference the QSK45/60 service tools in this document for the correct crankshaft seal installation tool.

The unitized seal provides improved durability over the previous conventional-design seals.

special instructions for K19/QSK19 Rear Crankshaft Seals

Crankshaft Standardization

A new high-wall crankshaft (material added in area of rod pin to make sidewalls higher), Part Number 3096362 (bare crankshaft), was released into production and service for both K19/QSK19 construction and marine engines in May 1997. The crankshaft was released for generator set engines in October 1997. The rear crankshaft seal area of the crankshaft, Part Number 3096362, is 0.020-inch larger in diameter, 152.908 mm versus 152.4 mm [6.020 inches versus 6 inches], than the crankshaft, Part Number 201082, previously used in the K19 engine. The increased diameter was required to accommodate the use of the unitized seal for the QSK19 engine.

When replacing the rear crankshaft seal on K19 engines built or upfitted in the field with crankshaft, Part Number 3096362 (identified by 0.254-mm [0.010-in] step between the flywheel pivot and seal area once the seal is removed). Use either the standard K19 service seal, Part Number 3870890 (dry), or unitized QSK19 seal, Part Number 3804704 (dry), can be used; however, the unitized seal is recommended because of improved seal durability.

When replacing the rear seal on wet-housing applications using crankshaft, Part Number 3096362, the service rear seal, Part Number 3090297, **must** be used on both K19 and QSK19 engines. This seal includes a wear sleeve that is designed to accommodate the 0.254-mm [0.010-in] step in the crankshaft seal area.

Rear Gear Drive Crankshaft

To date, all QSK19 rear gear drive engines have been built with the standard K19 rear gear drive crankshaft, Part Number 3016098 (bare crankshaft). When replacing the rear crankshaft seal on QSK19 engines built with the crankshaft, Part Number 3016098, the standard K19 rear seal, Part Number 3005886 (wet), or crankshaft, Part Number 3870890 (dry), **must** be used.

The crankshaft, Part Number 3016098, has been superseded by high-sidewall crankshaft, Part Number 3331587 (bare crankshaft), for both the K19 and QSK19 rear gear drive engines. The rear seal area of the crankshaft, Part Number 3331587, is 0.508-mm [0.020-in] larger in diameter, 152.908 mm versus 152.4 mm [6.020 inches versus 6.000 inches], than the crankshaft, Part Number 3016098, previously used on K19 and QSK19 rear gear drive engines.

The crankshaft, Part Number 3331587, can be identified once the rear seal is removed by the 0.254-mm [0.010-in] step between the flywheel pilot diameter and the seal area.

When replacing the rear crankshaft seal on the rear gear drive engines built with or upfitted with the high-wall crankshaft, Part Number 3331587, it is recommended that the QSK19 unitized rear seal, Part Number 3804704 (dry), be used because of improved seal durability.

When replacing the rear seal on **wet-housing** applications using the crankshaft, Part Number 3331587, service rear seal, Part Number 3090297 (nonunitized seal), **must** be used on both the K19 and QSK19 engines. This seal includes a wear sleeve that is designed to accommodate the 0.254-mm [0.010-in] step in the crankshaft seal area.

Seal Location	Producti on Seal Part Number	Service Seal with Installtio n Sleeve Part Number (if required)	Service Seal Assembl y with Wear Sleeve Part Number	Pose Excluder	Service Tool for Service Seal Part Number	Service Tool for Service Seal with Wear Sleeve Part Number	Repair Sleeve (Thin-Wall 0.012-Inch)
V378/504 Engines (Right-Hand Rotation)							
Front Cranksh aft Seal	212227	3010457	-	-	ST971	-	3010464
Rear Cranksh aft Seal (Dry)	3010458	3010459	-	-	ST972	-	3010465

Seal Location	Producti on Seal Part Number	Service Seal with Installtio n Sleeve Part Number (if required)	Service Seal Assembl y with Wear Sleeve Part Number	Pose Excluder	Service Tool for Service Seal Part Number	Service Tool for Service Seal with Wear Sleeve Part Number	Repair Sleeve (Thin-Wall 0.012-Inch)
Rear Cranksh aft Seal (Wet)	3275373	3008575	-	-	ST972	-	3010465
Accesso ry Drive	140896	106276	-	-	-	-	-
V378/504 Engines (Left-Hand Rotation)							
Front Cranksh aft Seal	3010462	3010461	-	-	ST971	-	3010464
Rear Cranksh aft Seal (Dry)	3010460	3010461	-	-	ST972	-	3010465
Rear Cranksh aft Seal (Wet)	-	-	-	-	-	-	-
Accesso ry Drive	210764	106276	-	-	-	-	-
V/VT555 Engine (Right-Hand Rotation)							
Front Cranksh aft Seal	3010460	3013461	-	-	ST1302	-	3010465
Rear Cranksh aft Seal (Dry)	3010458	3010459	-	-	ST1302	-	3010465
Rear Cranksh aft Seal (Wet)	3275373	3008575	-	-	-	-	3010465
Accesso ry Drive	210764	106276	-	-	-	-	-

Seal Location	Producti on Seal Part Number	Service Seal with Installtio n Sleeve Part Number (if required)	Service Seal Assembl y with Wear Sleeve Part Number	Pose Excluder	Service Tool for Service Seal Part Number	Service Tool for Service Seal with Wear Sleeve Part Number	Repair Sleeve (Thin-Wall 0.012-Inch)
V/VT555 Engine (Left-Hand Rotation)							
Front Cranksh aft Seal	3010458	3010459	-	-	ST1302	-	3010465
Rear Cranksh aft Seal (Dry)	3010460	3010461	-	-	ST1302	-	3010465
Rear Cranksh aft Seal (Wet)	-	-	-	-	-	-	-
Accesso ry Drive	210764	106276	-	-	-	-	-
VT-8-370 Marine (Left-Hand Rotation)							
Front Cranksh aft Seal (Left-Hand Rotation)	3021093	3021094	-	-	ST1071	-	3006740
Rear Cranksh aft Seal (Dry)	3013347	3013347	-	-	ST1108	-	3006741
V/VT903 Engines							
Front Cranksh aft Seal	3020191	3020184	-	-	ST1071	-	3006740
Rear Cranksh aft Seal (Dry)	218539	218539	-	-	ST1108	-	3006741

Seal Location	Producti on Seal Part Number	Service Seal with Installtio n Sleeve Part Number (if required)	Service Seal Assembl y with Wear Sleeve Part Number	Pose Excluder	Service Tool for Service Seal Part Number	Service Tool for Service Seal with Wear Sleeve Part Number	Repair Sleeve (Thin-Wall 0.012-Inch)
Rear Cranksh aft Seal (Wet)	3002356	3002356	-	-	3375818	-	3006741
Accesso ry Drive	3020187	3020187	-	-	-	-	-
VT903 Bus Engines (Left-Hand Rotation)							
Front Cranksh aft Seal	3021093	3021094	-	-	ST1071	-	3006740
Rear Cranksh aft Seal (Dry)	3016660	3016660	-	-	ST1108	-	3006741
Rear Cranksh aft Seal (Wet)	-	-	-	-	-	-	-
Accesso ry Drive	-	-	-	-	-	-	-
K19 Engines							
Front Cranksh aft Seal	3016787	3016792 *	3803994	3331663	3824760	3824760	-
Rear Cranksh aft Seal (Dry)	4922217	3870890	3099421	-	-	3824971	-
Rear Cranksh aft Seal (Wet)	3005883	3005886	3090298	-	-	3824971	-

[illegible]

Seal Location	Producti on Seal Part Number	Service Seal with Installtio n Sleeve Part Number (if required)	Service Seal Assembl y with Wear Sleeve Part Number	Pose Excluder	Service Tool for Service Seal Part Number	Service Tool for Service Seal with Wear Sleeve Part Number	Repair Sleeve (Thin-Wall 0.012-Inch)
Front Cranksh aft Seal	3020189	3020182	-	-	ST1304	-	3006744
Rear Cranksh aft Seal (Dry)	3082142	3630681	3070080	-	ST1093	3824424	-
Rear Cranksh aft Seal (Wet)	3001771	3005887	3076066	-	ST1093	3824424	-
Accesso ry Drive(Fu el Pump Side)	3020187	3020187	-	-	3825104	-	-
Accesso ry Drive (Water Pump Side)	3020190	3020190	-	-	3825104	-	-
Generato r Drive Housing	3020188	3020188	-	-	3825104	-	-
V28 Engines (Left-Hand Rotation)							
Front Cranksh aft Seal	3021095	3021096	-	-	ST1304	ST1304	-
Rear Cranksh aft Seal (Dry)	3012542	3012544	-	-	ST1093	ST1093	-

Seal Location	Producti on Seal Part Number	Service Seal with Installtio n Sleeve Part Number (if required)	Service Seal Assembl y with Wear Sleeve Part Number	Pose Excluder	Service Tool for Service Seal Part Number	Service Tool for Service Seal with Wear Sleeve Part Number	Repair Sleeve (Thin-Wall 0.012-Inch)
Rear Cranksh aft Seal (Wet)	-	-	-	-	ST1093	-	-
Accesso ry Drive(Fu el-Pump Side)	3004316	3004316	-	-	3825104	-	-
Accesso ry Drive (Water-Pump Side)	3020188	3020188	-	-	3825104	-	-
Generato r Drive Housing	3020190	3020190	-	-	3825104	-	-
QST30 Engines							
Front Cranksh aft Seal	3016787	3016792	3803994	-	3163485	3163485	-
Rear Cranksh aft Seal (Dry)	3092821	3099190	3099199	-	3825191	3825191	-
Rear Cranksh aft Seal (Wet)	3092624	3099422	3099200	-	3825191 and 4918762	3825191 and 4918762	-
Accesso ry Drive	3092812	3099197	-	-	-	-	-

Seal Location	Producti on Seal Part Number	Service Seal with Installtio n Sleeve Part Number (if required)	Service Seal Assembl y with Wear Sleeve Part Number	Pose Excluder	Service Tool for Service Seal Part Number	Service Tool for Service Seal with Wear Sleeve Part Number	Repair Sleeve (Thin-Wall 0.012-Inch)
Accesso ry Drive (Water-Pump Side)	3092366	3099198	-	-	-	-	-
Fuel-Pump Drive	3093304	3099196	-	-	-	-	-
QSK38/50 and K38/50 Engines							
Front Cranksh aft Seal	3016787	3016792 *	-	-	5298874	-	-
Rear Cranksh aft Seal (Dry)	4396606	3642365	3642365	-	ST1093	3824424	-
Rear Cranksh aft Seal (Wet)	3001771	3005887	3076066	-	ST1093	3824424	-
Accesso ry Drive(Fu el-Pump Side)	5694571	5573466	-	-	6378385	6378385	-
Accesso ry Drive(Wa ter-Pump Side)	5694571	5573466	-	-	6378385	6378385	-
QSK45/60 Engines							
Front Cranksh aft Seal	3171910	3171910	Not Required	Not Required	Not Required	Not Required	Not Required

Seal Location	Producti on Seal Part Number	Service Seal with Installtio n Sleeve Part Number (if required)	Service Seal Assembl y with Wear Sleeve Part Number	Pose Excluder	Service Tool for Service Seal Part Number	Service Tool for Service Seal with Wear Sleeve Part Number	Repair Sleeve (Thin-Wall 0.012-Inch)
Rear Cranksh aft Seal (Dry)	3171911	3171911	Not Required	Not Required	Not Required	Not Required	Not Required
Rear Cranksh aft Seal (Wet)	-	-	Not Required	Not Required	Not Required	Not Required	Not Required
Accesso ry Drive(Fu el-Pump Side)	3171909	3171909	Not Required	Not Required	Not Required	Not Required	Not Required
Accesso ry Drive(Wa ter-Pump Side)	3171909	3171909	Not Required	Not Required	Not Required	Not Required	Not Required
*See Technical Service Bulletin, Front Crankshaft Seal with Wear Sleeve and Press-On Shaft Excluder (POSE), TSB120221							

Document History

Date	Details
2010-11-3	Module Created
2013-11-25	none
2013-12-3	outfile
2018-2-2	Added note
2018-2-6	Revised note
2018-2-8	Revised note

Date	Details
2021-2-2	Revised the listed part number for Rear Crankshaft Seal (Dry) for the K19 Engines from 3160925 to 4922217.
2023-5-5	Updated part numbers.
2024-1-9	Updated Table.

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